

Krishna Teja Sunkara

Engineering Lead | Distributed Systems | GenAI Platforms | AWS
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Bengaluru, India | Open to Relocation

Professional Summary

Principal-level Engineering Leader with **12+** years of experience architecting large-scale distributed systems and GenAI platforms on AWS. Spearheaded Delta Trace, an organization-wide LLM-powered observability platform (AWS Bedrock), reducing MTTR by 30% and standardizing system visibility across teams. Defined and drove adoption of resiliency standards including automated health checks and cross-region failover, improving reliability across critical services. Proven track record of influencing cross-team architecture, optimizing cloud costs (~25%), and building highly available systems (99.9% uptime) at enterprise scale.

Key Skills

Programming Languages: Java 17+, Python, SQL, TypeScript

Frameworks: Spring Boot, Spring, Angular (12+), FastAPI

Distributed Systems & Architecture: Microservices, Event-Driven Architecture, Serverless Systems, Resiliency Patterns, Observability, System Design

Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, DynamoDB, ECS, SQS, CloudWatch, CloudFront, Route 53, CodePipeline, CloudFormation, EventBridge), Docker, GitLab CI/CD, Terraform

Databases: DynamoDB, MySQL, MongoDB

Engineering Practices: TDD, Agile (Scrum), Performance Optimization, Security (SAST/DAST/SCA)

AI / GenAI: Amazon Bedrock, Nova Pro, Anthropic Claude models, AI Agents, Prompt Engineering, AWS Textract, Cursor, AWS KIRO, Microsoft Copilot

Professional Experience

Lead Developer – Delta Airlines | Jun 2023 – Present

- Spearheaded “**Delta Trace**”, an organization-wide GenAI-powered observability platform leveraging LLMs (Amazon Bedrock – Nova), enabling automated system understanding across AWS, ROSA, and on-prem ecosystems, reducing **MTTR by ~30%** and eliminating tribal knowledge dependencies
- Drove adoption of **Delta Trace across 10+ teams and 50+ services**, establishing it as the standard platform for **system visibility, dependency mapping, and incident diagnosis** across the organization
- Architected an **AI-driven discovery engine** analyzing **100+** repositories and distributed infrastructure (repository scanning, infrastructure parsing, drift detection), generating real-time dependency graphs for **blast-radius** analysis and failure prediction
- Designed **cost-efficient LLM architecture using model tiering** (Nova for high-volume workloads, Claude for complex reasoning), optimizing **latency, accuracy, and inference cost** across use cases
- Automated system documentation and dependency mapping, reducing **manual effort by ~40%** and eliminating outdated architecture diagrams
- Defined organization-wide resiliency standards (automated health checks, cross-region failover), ensuring **99.9% uptime**, and served as the technical authority for observability and resiliency across distributed systems
- Influenced **architecture decisions across teams**, driving consistency in observability, resiliency patterns, and distributed system design
- Led a **cross-functional team of 10 engineers across US–India**, mentoring engineers and driving delivery of mission-critical distributed APIs
- Reduced cloud spend by **~25%** through FinOps initiatives (DynamoDB TTL, network optimization, rightsizing and scheduling)

- Improved security posture, resolving **100+ vulnerabilities** by integrating **Veracode SAST/DAST/SCA** into CI/CD, enabling early detection
- Reduced API failures by **~30%** by migrating on-prem services to AWS (API Gateway, Lambda) with modernized integrations
- Mentored engineers through regular 1:1s, improving team performance and strengthening engineering standards
- Led technical interviews and influenced hiring decisions, contributing to building high-performing teams and strengthening engineering standards

Technical Lead (Full Stack) – London Stock Exchange Group (LSEG) | *Jul 2022 – Jun 2023*

- Led a **team of 10 engineers** delivering Angular and Spring Boot applications in a globally distributed environment
- Improved real-time dashboard performance by **~50%** through frontend optimization and efficient data handling (RxJS)
- Implemented GitLab CI/CD pipelines, reducing manual release efforts by **~40%** and enabling automated deployments

Senior Full Stack Engineer – Taylor & Francis Group | *Jul 2021 – Jul 2022*

- Developed **Spring Boot REST APIs** and integrated with publishing microservices for metadata, journal, and article transfer
- Implemented **OAuth 2.0** authentication, securing both user and service-to-service communications
- Improved system reliability by increasing **test coverage to 90%**

Senior Software Engineer – Refinitiv (acquired by LSEG) | *Apr 2018 – Jul 2021*

- Built high-performance, multi-threaded Java applications for real-time market data processing at scale
- Introduced **Docker**, reducing local setup time by **~20%** and simplifying onboarding
- Implemented **JWT-based authentication** strengthening security for web applications
- Built a multi-threaded REST client for parallel TRIT API calls, delivering **~10x faster** document processing

QA Engineer / Senior QA Engineer – Thomson Reuters (business unit later became Refinitiv) | *Apr 2014 – Apr 2018*

- Achieved **100% automation** for SDI and database comparison testing using Java, reduced regression cycle time by **60%** using Selenium

Selected System Impact

- **Forest Spirit (LSEG):** Developed **AI-driven** document processing pipelines, including a custom PDF-to-HTML transformation system enabling paragraph-level content extraction and analysis for ML workflows
- **Fast Tick (Refinitiv Dev):** Designed distributed data processing pipelines handling high-frequency, real-time and multi-GB datasets, improving processing throughput by **~10x** using concurrency patterns. Powered downstream systems including Refinitiv Eikon for real-time market data and analytics

Certifications

- AWS Certified AI Practitioner
- AWS Certified Developer – Associate
- AWS Certified Solutions Architect – Associate

Education

- **Bachelor of Technology (B.Tech), Computer Science & Engineering** – PVPSIT, Vijayawada (2008 – 2012)